

Sentiment Classifier : : CHEAT SHEET



About

Kaiaulu provides support for the download of comment data from multiple sources e.g. GitHub, Jira, and mailing lists (see refresher-cheatsheet.pdf). This cheatsheet covers how to add sentiment labels to each comment by using machine learning classifiers for sentiment analysis.

Project Config Setup

Before running a vignette, you must set up your project configuration file (examples in conf/) to specify the paths to download the data.

Required Fields

- **issue_tracker:**
 - github:
 - project_key_1:
 - owner:
 - repo:
 - issue_or_pr_comment:
 - issue_event:
 - commit:
 - commit_comments:
 - pr_comments:
 - jira:
 - project_key_1:
 - issue_comments:
- **mailing_list:**
 - mod_mbox:
 - project_key_1:
 - pipemail:
 - project_key_1:

You should also specify the required external tools in tools.yml:

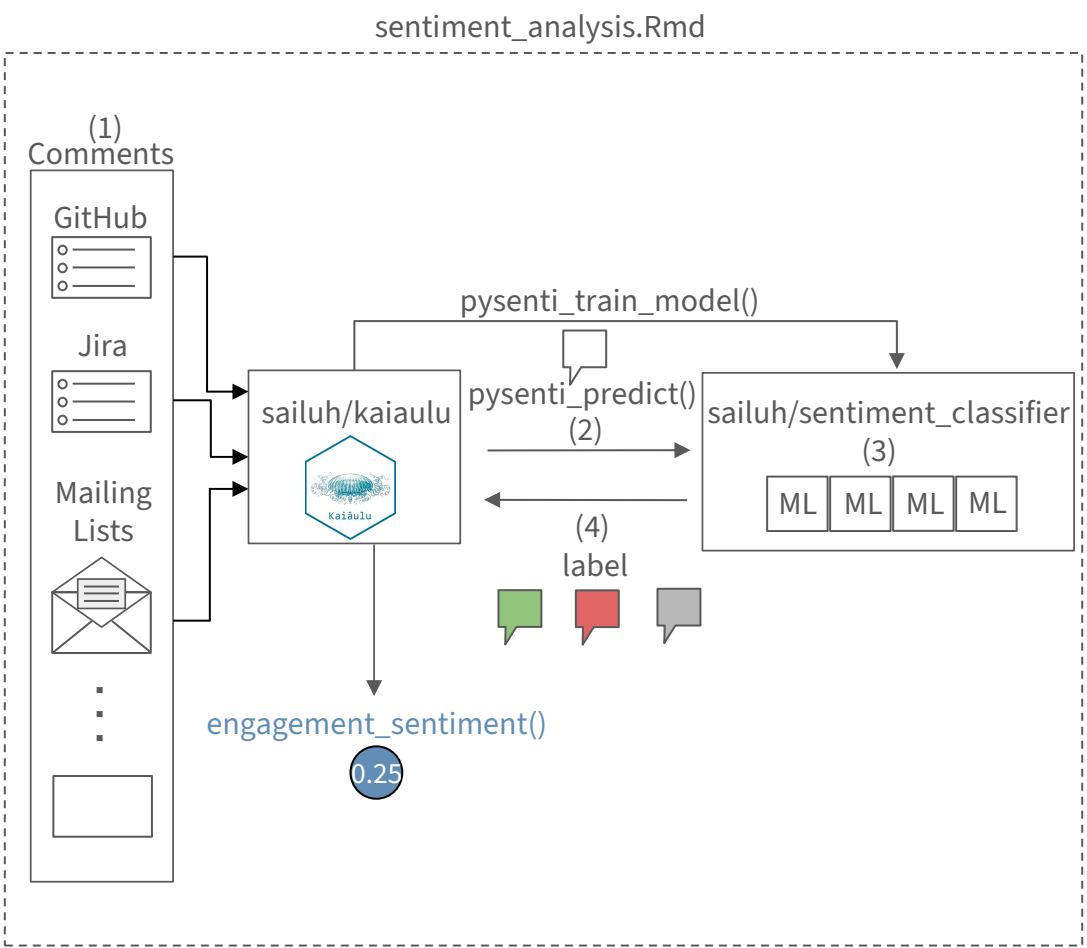
Tools Config Setup

Required Fields

- **pysenti:**
 - ..\..\sentiment_classifier\exec\train_or_predict.py
- **perceval:**

Comment Sentiment

Kaiaulu’s sentiment_analysis.Rmd interfaces with a family of sentiment classifier models. The classifier leverages comment data (1) obtainable by Kaiaulu downloaders.



Kaiaulu passes comments (1) to the sentiment classifier using pysenti_predict() (2). The tools.yml path for the sentiment classifier follows the same pattern as the other third-party tools.

The sentiment classifier labels each comment as **positive** (green), **negative** (red), or **neutral** (grey) using machine learning models (3) and returns that label (4) to Kaiaulu.

The sentiment labels can also be used as a measure of **project emotional engagement**.

Related Vignettes

See supplementary notebooks that go in-depth on downloading and parsing communication data and conducting the full sentiment analysis workflow.

1. [download_github_events.Rmd](#)
2. [download_github_pull_requests.Rmd](#)
3. [sentiment_analysis.Rmd](#)

Engagement Metrics

Kaiaulu supports time-series analysis of developer comments across all sources via various engagement metrics. For two of them, developer’s comments and sentiments are grouped respectively by time step within a moving window to form a time series of engagement.

1. **engagement_communication()**
Counts the overall number of messages a developer sent.
2. **engagement_sentiment()**
Produces an engagement sentiment score.

